SOFTWARE REQUIREMENT ENGINEERING

Final SRS

Title:

" QuickBite – Online Food Delivery System "

Submitted to:

Ma’am Sanam Shoukat

Submitted by:

Alisha Kanwal (FA-23/BSSE/165-D)

Alina Irshad (FA-23/BSSE/199-D)

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Department of Software Engineering

Lahore Garrison University

Team Members and Roles

Team Member	Role	Responsibilities
Alina Irshad	Requirements Analyst	D1 (Project Proposal), D2 (Requirements Elicitation Report), UML diagrams
Alisha Kanwal	Project Lead	D3 (SRS Draft), prototype (figma), Final SRS + Presentation D4

Revision History

Version	Date	Description	Author
1.0	May 2026	Initial Draft of SRS Document	Project Team

DOCUMENT APPROVAL

This Software Requirements Specification (SRS) document for the "QuickBite – Online Food Delivery System" has been prepared as part of the Software Requirement Engineering Lab semester project.

The document contains the functional requirements, non-functional requirements, UML diagrams, and data requirements of the proposed system.

Table of Contents

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (**SRS**) document is to define the functional and non-functional requirements of the "**QuickBite** – Online Food Delivery System". This document describes the features, system behavior, user interactions, and constraints of the proposed system.

The SRS document serves as a communication medium between developers, stakeholders, designers, testers, and project team members. It provides a clear understanding of the system requirements and acts as a reference during system design, development, testing, and maintenance phases.

The **QuickBite** system aims to provide a simple, efficient, and user-friendly online food ordering and delivery platform that connects customers, restaurants, delivery riders, and administrators through a centralized digital system.

1.2 Scope

QuickBite is an online food delivery system designed to simplify the food ordering and delivery process. The system will allow customers to browse restaurants, search for food items, place orders, make payments, and track deliveries in real time.

Restaurant owners will be able to manage menus, accept or reject customer orders, and update food preparation status. Delivery riders will receive delivery assignments and update delivery progress. Administrators will monitor the entire system and manage users, restaurants, and reports.

The system will provide:

- User registration and login
- Restaurant browsing and menu viewing
- Food search and filtering

- Cart management
- Online payment and cash on delivery
- Order placement and cancellation
- Real-time order tracking
- Delivery rider assignment
- Ratings and reviews
- Administrative management features

The system will focus on usability, fast performance, secure transactions, and efficient order management.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
SRS	Software Requirements Specification
UML	Unified Modeling Language
ERD	Entity Relationship Diagram
Admin	System Administrator responsible for system management
Customer	User who orders food through the system
Rider	Delivery person responsible for delivering food
GUI	Graphical User Interface
COD	Cash on Delivery
Database	Organized collection of system data

1.4 References

- Software Requirement Engineering Lab Guidelines
- IEEE 830 Software Requirements Specification Standard
- Research articles related to online food delivery systems

- User interview and survey responses collected during elicitation phase

1.5 Document Overview

This document is organized into multiple sections describing different aspects of the QuickBite Online Food Delivery System.

- **Section 1** provides an introduction to the system including purpose, scope, definitions, and references.
- **Section 2** explains the overall description of the system, product functions, user characteristics, operating environment, and system constraints.
- **Section 3** contains the detailed functional requirements, non-functional requirements, and use case specifications of the system.
- **Section 4** includes UML diagrams such as the use case diagram and activity diagrams.
- **Section 5** explains the data requirements including the Entity Relationship Diagram (ERD) and data dictionary.
- **Section 6** contains appendices and additional supporting information related to the project.

2. OVERALL DESCRIPTION

2.1 Product Perspective

QuickBite is a standalone online food delivery system which connects customers, restaurants, delivery riders, and administrators on a single platform. The system will work as a centralized platform where all food ordering and delivery activities are managed digitally.This system replaces manual food ordering methods and improves communication, speed, and accuracy of orders.

2.2 Product Functions

The system will provide the following main functions:

- User registration and login

- Restaurant browsing and food search
- Menu viewing with prices
- Cart management
- Order placement and cancellation
- Online payment and cash on delivery
- Real-time order tracking
- Delivery rider assignment
- Order status updates
- Ratings and reviews
- Admin panel for system control

2.3 User Classes and Characteristics

Customer

- Places food orders
- Uses mobile/web application
- Needs simple and fast interface

Restaurant Owner

- Manages menu items
- Accepts/rejects orders
- Updates order status

Delivery Rider

- Delivers food orders
- Updates delivery status

- Uses mobile app for navigation

System Administrator

- Manages users and restaurants
- Monitors system activity
- Handles complaints and reports

Payment Service Provider

- Processes online payments securely

2.4 Operating Environment

- Web-based application
- Mobile responsive interface
- Windows, Android, iOS compatible browsers
- Backend server with database support (MySQL/SQL Server)
- Internet connection required

2.5 Design and Implementation Constraints

- System must support real-time updates
- Secure authentication is required
- Limited budget (student project constraint)
- Must be developed using simple technologies (HTML, CSS, JS, database)
- Must follow modular design
- Time limitation for project completion

2.6 Assumptions and Dependencies

Assumptions:

- Users have basic internet access
- Restaurants will update their menu regularly
- Riders will use mobile devices

Dependencies:

- Internet connectivity
- Payment gateway services
- Database system availability
- Third-party APIs (if used for maps or payment)

3. SPECIFIC REQUIREMENTS

3.1 Functional Requirements

FR ID	Functional Requirement
FR-01	System shall allow users to register an account.
FR-02	System shall allow users to login securely.
FR-03	System shall allow password recovery.
FR-04	System shall display list of restaurants.
FR-05	System shall allow food search.
FR-06	System shall display restaurant menu.
FR-07	System shall allow add to cart functionality.
FR-08	System shall allow update/remove cart items.
FR-09	System shall allow order placement.
FR-10	System shall support online payment.

FR-11	System shall support cash on delivery.
FR-12	System shall send order confirmation notifications.
FR-13	System shall allow restaurant to accept/reject orders.
FR-14	System shall update order status.
FR-15	System shall assign delivery riders automatically.
FR-16	System shall allow riders to update delivery status.
FR-17	System shall allow order tracking.
FR-18	System shall maintain order history.
FR-19	System shall allow ratings and reviews.
FR-20	System shall allow admin to manage system data.

3.2 Non-Functional Requirements

NFR ID	Non-Functional Requirement
NFR-01	System response time should be 2–3 seconds.
NFR-02	System should ensure secure password encryption.
NFR-03	System should be user-friendly.
NFR-04	System should be available 24/7.
NFR-05	System should ensure data reliability.
NFR-06	System should be scalable for more users.
NFR-07	System should be compatible with mobile and web platforms.
NFR-08	System should be maintainable and easy to update.

3.3 Use Case Specifications

3.3.1 User Registration

Field	Description
Use Case Name	User Registration
Actor	Customer
Precondition	User is not registered
Postcondition	Account is created successfully

Main Flow	User enters details → System validates → Account created
Alternate Flow	If data invalid → error message shown

3.3.2 User Login

Field	Description
Use Case Name	User Login
Actor	Customer / Restaurant / Admin
Precondition	User must be registered
Postcondition	User accesses system
Main Flow	Enter credentials → System verifies → Login success
Alternate Flow	Wrong password → error message

3.3.3 Search Food Items

Field	Description
Use Case Name	Search Food Items
Actor	Customer
Precondition	User is logged in
Postcondition	Food items displayed
Main Flow	Enter search → System shows results
Alternate Flow	No result found → message shown

3.3.4 Add to Cart

Field	Description
Use Case Name	Add to Cart
Actor	Customer
Precondition	Food item selected
Postcondition	Item added to cart
Main Flow	Select item → Click add → System updates cart

Alternate Flow	Item unavailable → error shown

3.3.5 Place Order

Field	Description
Use Case Name	Place Order
Actor	Customer
Precondition	Cart must have items
Postcondition	Order is created
Main Flow	Checkout → Select payment → Confirm order
Alternate Flow	Payment failed → order not placed

3.3.6 Make Payment

Field	Description
Use Case Name	Make Payment
Actor	Customer
Precondition	Order placed
Postcondition	Payment completed
Main Flow	Select method → Pay → confirmation
Alternate Flow	Payment failure → retry

3.3.7 Track Order

Field	Description
Use Case Name	Track Order
Actor	Customer
Precondition	Order is placed
Postcondition	Status displayed
Main Flow	Open order → view live status
Alternate Flow	Order not found

3.3.8 Manage Restaurant Orders

Field	**Description**
Use Case Name	Manage Orders
Actor	Restaurant Owner
Precondition	Order received
Postcondition	Order status updated
Main Flow	Accept/reject → update status
Alternate Flow	Delay in response

3.3.9 Assign Delivery Rider

Field	Description
Use Case Name	Assign Rider
Actor	System Admin
Precondition	Order ready
Postcondition	Rider assigned
Main Flow	System selects rider → assigns order
Alternate Flow	No rider available

3.3.10 Manage Users

Field	Description
Use Case Name	Manage Users
Actor	Admin
Precondition	Admin logged in
Postcondition	User data updated
Main Flow	View users → update/delete
Alternate Flow	Invalid request

4. UML DIAGRAMS

4.1 Use Case Diagram

The Use Case Diagram of the QuickBite – Online Food Delivery System represents the interaction between different actors and the system. It shows the main functionalities provided by the system to its users.

The following actors are involved in the system:

- Customer
- Restaurant Owner
- Delivery Rider
- System Administrator
- Payment Service Provider

The diagram illustrates how each actor interacts with different system functions such as user registration, food ordering, payment processing, order management, and delivery tracking.

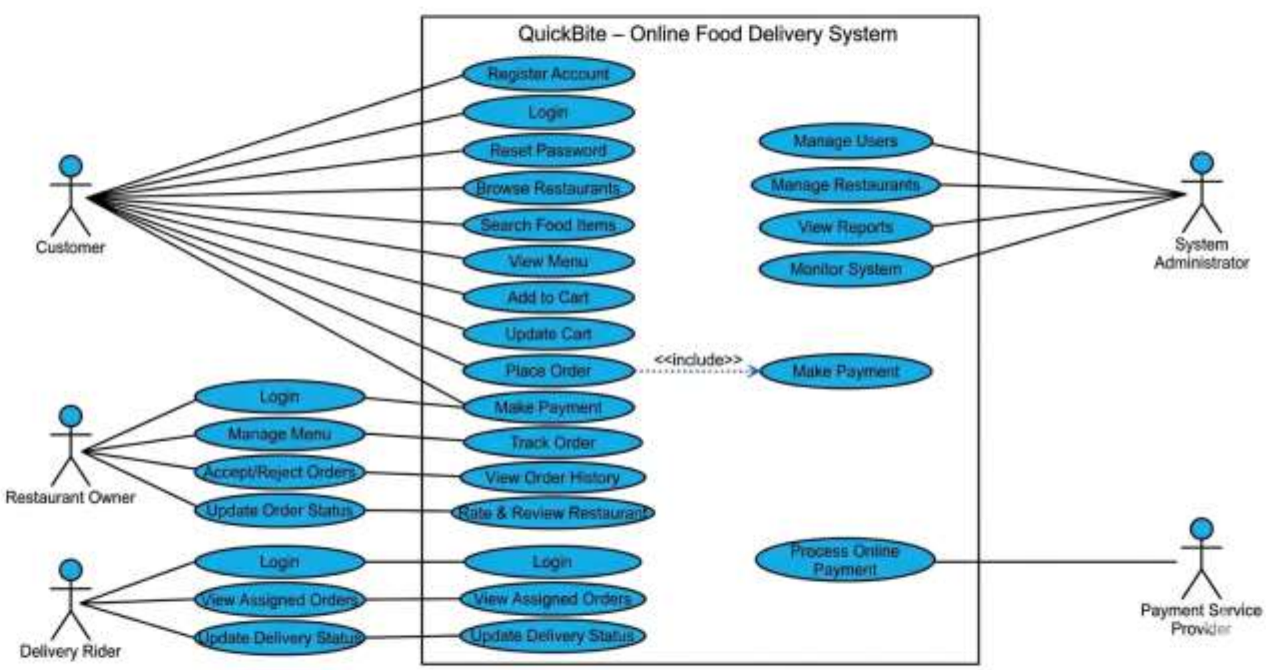

Figure 1Use Case Diagram of QuickBite System

4.2 Activity Diagram – User Registration

This activity diagram represents the process of user registration in the **QuickBite** system. The process begins when a user opens the application and selects the registration option.

The user enters personal details, which are validated by the system. If the information is valid, the system creates a new account and stores the data in the database. If invalid data is entered, the system displays an error message and asks the user to re-enter the information.

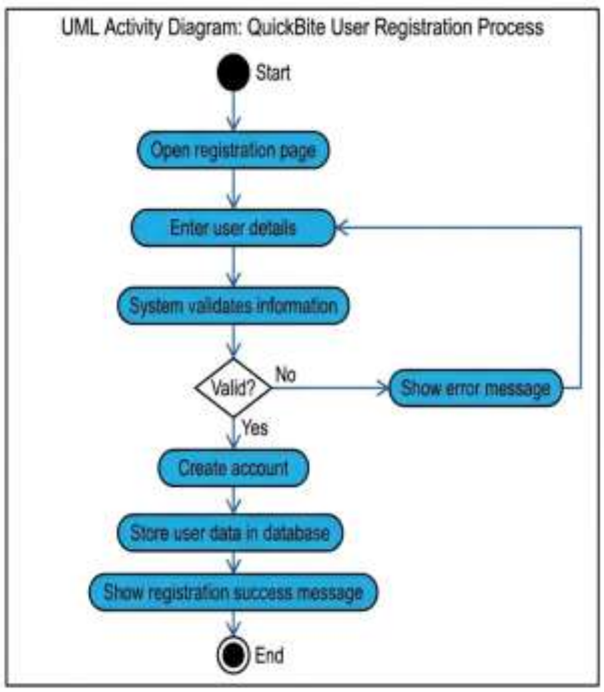

Figure 2 User Registration Activity Diagram

4.3 Activity Diagram – Place Order

This activity diagram describes the process of placing an order in the QuickBite system. The process starts when the customer logs into the system and browses available restaurants and food items.

The customer selects items, adds them to the cart, and proceeds to checkout. After selecting the payment method, the order is confirmed and sent to the restaurant for preparation. The system then updates the order status.

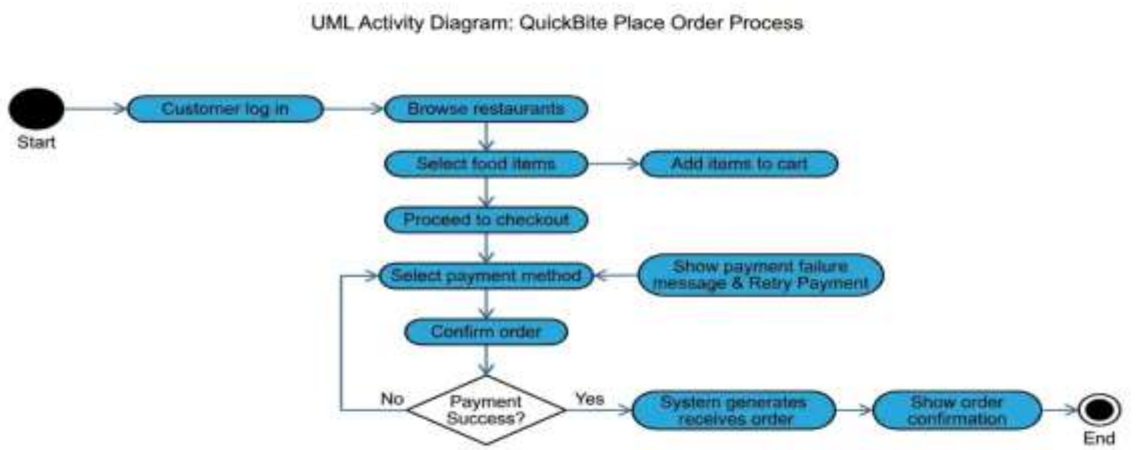

Figure 3 Place Order Activity Diagram

4.4 Activity Diagram – Order Delivery Process

This activity diagram represents the delivery process of an order. After the restaurant accepts the order and prepares the food, the system assigns a delivery rider.

The rider picks up the order and delivers it to the customer. During this process, the rider updates the delivery status, and the customer can track the order in real time until it is successfully delivered.

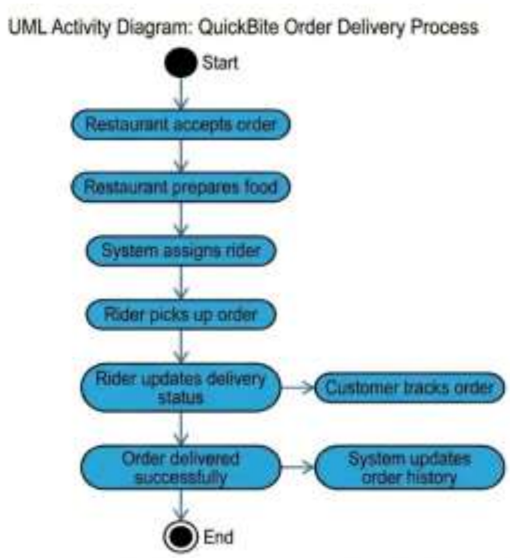

Figure 4Order Delivery Process Activity Diagram

4.5 Prototype (Figma UI Design)

The QuickBite Online Food Delivery System includes a user interface prototype designed using Figma. The purpose of this prototype is to visually represent the system's layout, user interaction flow, and overall design before actual development.

The prototype helps in understanding how users will interact with the system and ensures that the interface is simple, user-friendly, and efficient for all types of users including customers, restaurant owners, delivery riders, and administrators.

The following main screens have been designed in the Figma prototype:

- Login Page
- Registration Page
- Home Page (Restaurant Listing)
- Food Menu Page
- Payment Page
- Order Tracking Page
- Admin Dashboard

Figure 5 Admin dashboard

Figure 7 open app 1st page

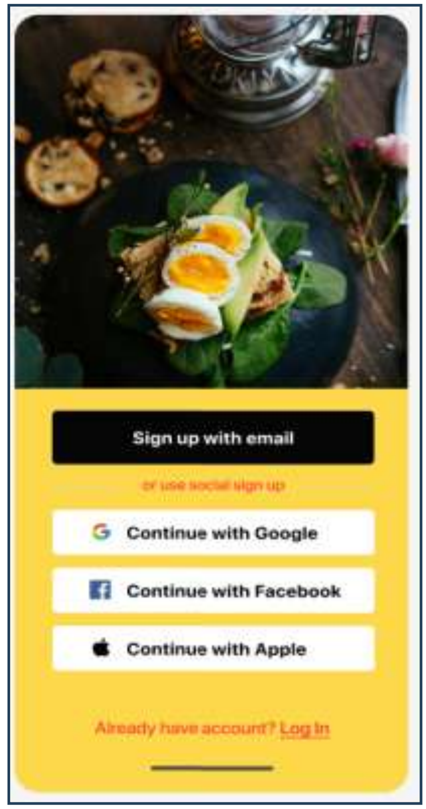

Figure 6 login page

Figure 8 home page

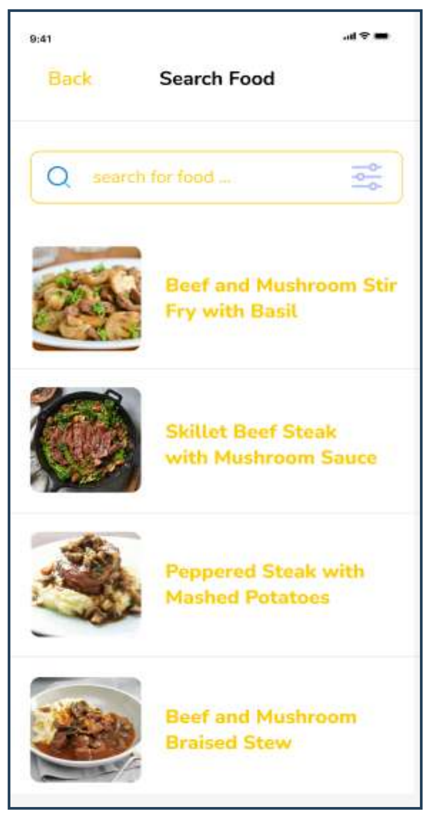

Figure 9 search food

Figure 11 payement method

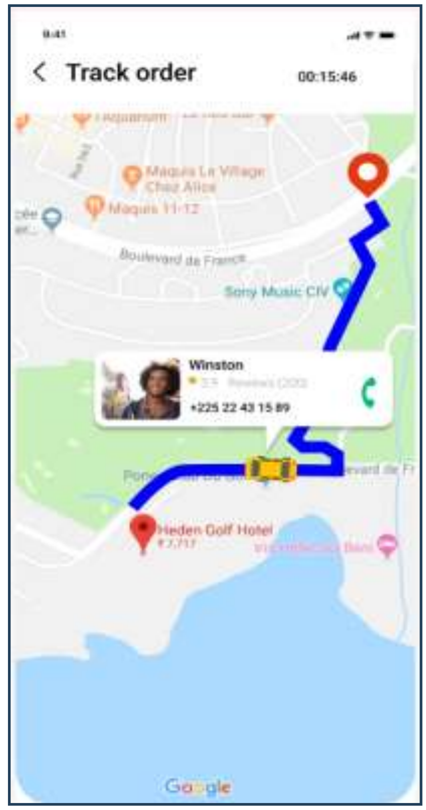

Figure 10 track order

5. DATA REQUIREMENTS

5.1 Entity Relationship Diagram (ERD)

The Entity Relationship Diagram (ERD) of the QuickBite system represents the structure of the database and the relationships between different entities.

The main entities of the system include:

- Customer
- Restaurant
- Menu Item
- Order

- Payment
- Delivery Rider
- Review

Relationships:

- A Customer can place multiple Orders.
- Each Order belongs to one Customer.
- A Restaurant contains multiple Menu Items.
- Each Order contains multiple Menu Items.
- Each Order is assigned to one Delivery Rider.
- Each Order has one Payment record.
- A Customer can write multiple Reviews for Restaurants.

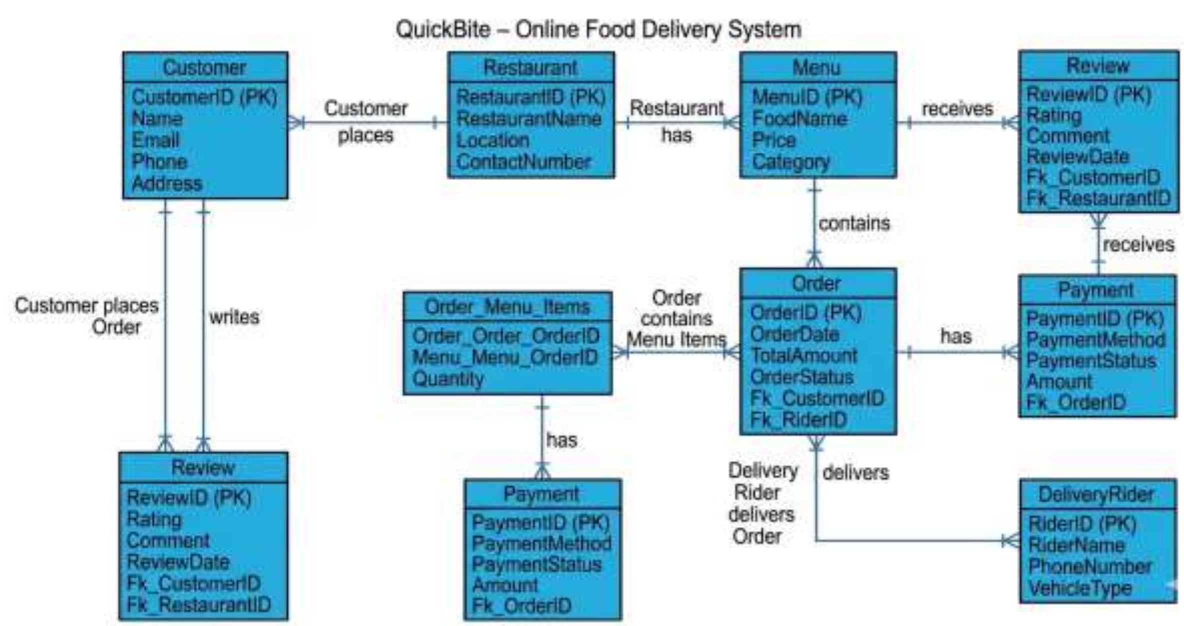

Figure 12Entity Relationship Diagram of QuickBite System

5.2 Data Dictionary

The data dictionary describes the structure of database tables, attributes, and their data types used in the system.

5.2.1 Customer Table

Attribute	Data Type	Description
Customer_ID	INT (PK)	Unique ID of customer
Name	VARCHAR	Customer full name
Email	VARCHAR	Email address
Password	VARCHAR	Encrypted password
Phone	VARCHAR	Contact number
Address	VARCHAR	Delivery address

5.2.2 Restaurant Table

Attribute	Data Type	Description
Restaurant_ID	INT (PK)	Unique restaurant ID
Name	VARCHAR	Restaurant name
Location	VARCHAR	Restaurant address
Contact	VARCHAR	Contact number

5.2.3 Menu Item Table

Attribute	Data Type	Description
Item_ID	INT (PK)	Unique food item ID
Restaurant_ID	INT (FK)	Linked restaurant
Item_Name	VARCHAR	Food item name
Price	DECIMAL	Item price
Category	VARCHAR	Food category

5.2.4 Order Table

Attribute	Data Type	Description
Order_ID	INT (PK)	Unique order ID
Customer_ID	INT (FK)	Who placed order
Restaurant_ID	INT (FK)	Restaurant involved
Order_Date	DATE	Order date
Status	VARCHAR	Pending/Delivered

5.2.5 Payment Table

Attribute	Data Type	Description
Payment_ID	INT (PK)	Unique payment ID
Order_ID	INT (FK)	Related order
Payment_Method	VARCHAR	COD / Online
Payment_Status	VARCHAR	Paid / Unpaid

5.2.6 Delivery Rider Table

Attribute	Data Type	Description
Rider_ID	INT (PK)	Unique rider ID
Name	VARCHAR	Rider name
Phone	VARCHAR	Contact number
Availability	VARCHAR	Available / Busy

5.2.7 Review Table

Attribute	Data Type	Description
Review_ID	INT (PK)	Unique review ID
Customer_ID	INT (FK)	Reviewer
Restaurant_ID	INT (FK)	Reviewed restaurant
Rating	INT	Star rating

Comment	TEXT	Feedback

6. Appendices

6.1 Stakeholder List

The following stakeholders are involved in the QuickBite – Online Food Delivery System:

Stakeholder	Role in the System
Customer	Places food orders and tracks deliveries
Restaurant Owner	Manages menu items and customer orders
Delivery Rider	Delivers food orders to customers
System Administrator	Manages users, restaurants, and system activities
Payment Service Provider	Handles secure online payment processing
Customer Support Staff	Resolves customer complaints and support issues

6.2 Requirement Priorities

The system requirements are categorized into different priority levels based on their importance and impact on system functionality.

Priority Level	Requirements
High Priority	User login, order placement, payment processing, order tracking
Medium Priority	Notifications, ratings and reviews, order history
Low Priority	Promotional offers, advanced search filters

6.3 Abbreviations

Abbreviation	Meaning
SRS	Software Requirements Specification
UML	Unified Modeling Language

ERD	Entity Relationship Diagram
GUI	Graphical User Interface
COD	Cash on Delivery
Admin	System Administrator

6.4 Future Enhancements

The following features may be included in future versions of the **QuickBite** system:

- AI-based food recommendation system
- Multi-language support
- Live GPS rider tracking
- Promotional discounts and coupon system
- Voice-based food search
- International food delivery support
- Advanced analytics and reporting dashboard

6.5 Conclusion

The **QuickBite** – Online Food Delivery System is designed to provide a simple, reliable, and user-friendly platform for online food ordering and delivery management. The system aims to improve communication between customers, restaurants, delivery riders, and administrators while reducing delays and manual management issues.

This Software Requirements Specification (**SRS**) document defines the functional and non-functional requirements, system behavior, UML diagrams, and data requirements of the proposed system. The document will serve as a reference during the design, development, testing, and maintenance phases of the project.